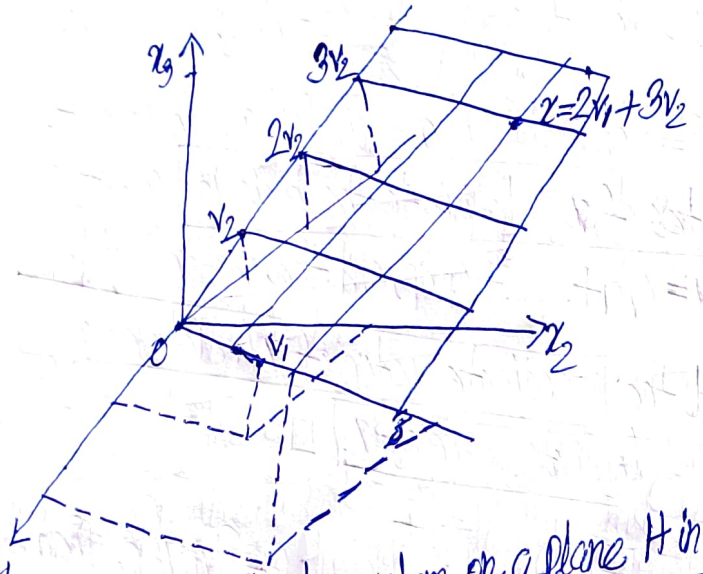



Figure 1: A coordinate system on a plane H in $R^3$

Notice that although points in H are also in $R^3$, they are completely determined by their coordinate vectors, which belong to $R^2$. The grid on the plane in Figure 1 makes H "look" like $R^2$.

The correspondence $x \mapsto [x]_B$ is a one-to-one correspondence between H and $R^2$ that preserves linear combinations. We call such a correspondence an **isomorphism**, and we say that H is **isomorphic** to $R^2$.

## The Dimension of a Subspace

By Exercise 27 and 28, it will be shown that if a subspace H has a basis of p vectors, then every basis of H must consist of exactly p vectors.

**Definition:** The **dimension** of a nonzero subspace H, denoted by dim H, is the number of vectors in any basis for H. The dimension of the zero subspace $\{0\}$ is defined to be zero.

The space $R^n$ has dimension n. Every basis for $R^n$ consists of n vectors.

A plane through 0 in $R^3$ is two-dimensional, and a line through 0 is one-dimensional.

**Example 2:** By Example 6 from Section 2.8, the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

has the basis of the Null-Space:

$$\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$